Tabular Foundation Models for Data Science Agents

Building a Data Science Agent

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DeepLearning.AI

LLMs vs TFMs

LLMs

Tasks over **unstructured** data,
no matter the domain

OpenAI ANTHROPIC MISTRAL AI_ cohere

- Excel at *language* & *knowledge*
- Generate *summaries, explanations, code*
- **Boost business productivity**

Tabular Foundation Models (TFMs)

Tasks over **structured** data,
no matter the domain

databases CSVs spreadsheets time series

PRIOR

- Excel at *numbers, statistics & patterns*
- Generate *predictions, forecasts, scores*
- **Directly plug into existing ML value chain**

LLMs for reasoning - not prediction

LLMs excel at language tasks: **parsing, lookup, extraction**.

CUSTOMERS.CSV

customer	tenure	plan	$/mo	tickets	churned
C-1041	34	annual	49	0	no
C-1042	3	monthly	89	4	**yes**
C-1043	18	monthly	59	1	no
C-1044	7	monthly	99	5	yes
C-1045	52	annual	39	0	no

USER

Did customer C-1042 churn?

LLM

Yes — C-1042 churned after 3 months on the monthly plan, with 4 support tickets.

✓ The answer was **in the data**.

What about unseen data?

The answer isn't in the table. The model **produces one anyway**.

CUSTOMERS.CSV

customer	tenure	plan	$/mo	tickets	churned
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C-1042	3	monthly	89	4	yes
C-1043	18	monthly	59	1	no
C-1044	7	monthly	99	5	yes
C-1045	52	annual	39	0	no
C-1046	5	monthly	94	3	?

USER

Will customer C-1046 churn?

LLM

Similar to other short-tenure monthly customers — I'd say **about 70% likely**.

! Plausible — but **nothing was *statistically* learned** from the rows. “70%” is a phrase, not a probability.

This is a prediction problem

Five customers with a known outcome, one without: **supervised learning.**

features **target**

tenure	plan	$/mo	tickets	churned
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7	monthly	99	5	yes
52	annual	39	0	no
5	monthly	94	3	?

training (context) set

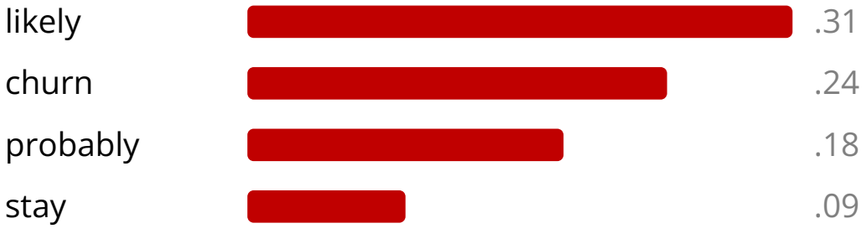
query

Every pattern the answer should be based on lives in the **labelled rows** . We need a model that actually **learns from them** .

Prediction is a different job

WHAT AN LLM IS TRAINED TO DO

This customer will ___



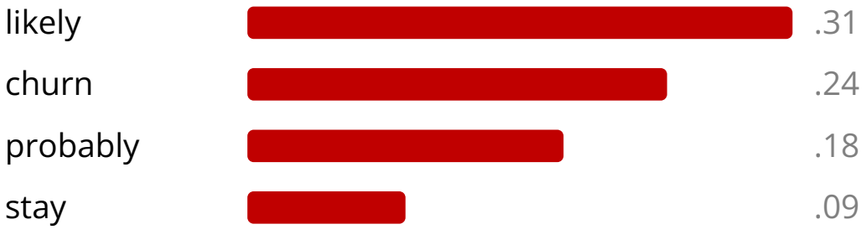
Trained to generate the next best token

A distribution over **words** → great to **reason** and **plan**

Prediction is a different job

WHAT AN LLM IS TRAINED TO DO

This customer will ___

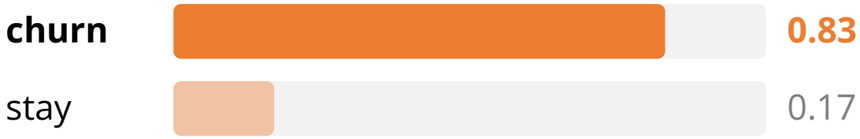


Trained to generate the next best token

A distribution over **words** → great to **reason** and **plan**

WHAT A PREDICTION TASK ASKS FOR

tenure=5 · monthly · $94 · 3 tickets → **P(churn | features) = ?**



A distribution over **outcomes**, calibrated on the rows → needed to **predict** and decide based on your data

TabPFN as a tool call

A pre-trained transformer that predicts in-context. **Your table is the context.**



Classical ML

churn data → churn model

prices → price model

risk → risk model

trains **one model per dataset**

TabPFN

churn · prices · risk

↓ read in-context by

one pre-trained model

trained **once**, no training loop

TabPFN as a tool call

USER Will customer C-1046 churn?

LLM agent

"This is a prediction task — call the tool, don't guess."

reasons

① tool call

tabpfn.predict (train = customers.csv, target = "churned", task = "classification", query = {tenure: 5, plan: "monthly", usd: 94, tickets: 3})

② result

{ "churn": **0.83**, "stay": 0.17 }

TabPFN tool

Learns about past customers in-context — one forward pass.

predicts

③ agent reasons over the result

C-1046 has an **83% churn probability** , learned from your past customers. Worth a retention offer.

TabPFN MCP

The **tabpfn MCP server** exposes 5 tools — a chain of ids from files to predictions.

local files → train_set_upload_id → fitted_train_set_id → test_set_upload_id → **prediction**

```
prepare_train_set_upload()
fit(upload_id, task)
get_fit_status(fitted id)
prepare_test_set_upload(fitted id)
predict(…, task)
```

- One fit, unlimited predicts — **fits get cached**